Code-Layout, Readability and Reusability

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Name	Rising Waters (Flood Prediction System)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Proper formatting followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for dataset, model and app
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive function names
5	Code Comments	Inline and block comments explain log c	Yes	Important sections documented
6	Modular Design	Code is split into reusable functions/modules	Yes	Functions divided into modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python	Model Training	High

2	Feature Engineering	Python	Prediction Modul	High
3	Flood Prediction Model	Python	Prediction System	High
4	Visualization Module	Python	Dashboard	High
5	Alert Module	Python	Risk Notification	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to understand
Reusability	5	Reusable modules
Documentation / Comments	5	Adequate comments
Overall Score	5	Good quality code